PROPERNESS OVER THE BASE OF A CLOSED SUBSET IS INDEPENDENT OF THE CLOSED SUBSPACE STRUCTURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete: the three implications are all carried out, with the union trick doing the work

Editorial note. The label lemma-closed-proper-over-base names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- reduced induced closed subspace structures and scheme-theoretic unions of closed subspaces
- properness, separatedness and quasi-compactness of morphisms of algebraic spaces and their permanence under closed immersions, compositions and images of universally closed maps
- bijective closed immersions as universal homeomorphisms

Source reference. Original definition 0CYZ: sources/stacks-chow/tags.txt, source lines 16259–16259. Prerequisite pointer invoked at proof line(s) 1057; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04XZ: sources/stacks-chow/tags.txt, source lines 6412–6412. Prerequisite pointer invoked at proof line(s) 1064; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03XG: sources/stacks-chow/tags.txt, source lines 5147–5147. Prerequisite pointer invoked at proof line(s) 1076; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04XY: sources/stacks-chow/tags.txt, source lines 6411–6411. Prerequisite pointer invoked at proof line(s) 1065; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08AJ: sources/stacks-chow/tags.txt, source lines 10534–10534. Prerequisite pointer invoked at proof line(s) 1081; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 05Z2: sources/stacks-chow/tags.txt, source lines 7676–7676. Prerequisite pointer invoked at proof line(s) 1071; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06ED: sources/stacks-chow/tags.txt, source lines 8212–8212. Prerequisite pointer invoked at proof line(s) 1075; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CZ1: sources/stacks-chow/tags.txt, source lines 16261–16261. Prerequisite pointer invoked at proof line(s) 1060; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 047X: sources/stacks-chow/tags.txt, source lines 5524–5524. Prerequisite pointer invoked at proof line(s) 1040; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite type. Let $T \subset |X|$ be a closed subset. The following are equivalent*

1. *the morphism $Z \to Y$ is proper if Z is the reduced induced algebraic space structure on T (Properties of Spaces, Definition 047X),*
2. *for some closed subspace $Z \subset X$ with $|Z| = T$ the morphism $Z \to Y$ is proper, and*
3. *for any closed subspace $Z \subset X$ with $|Z| = T$ the morphism $Z \to Y$ is proper.*

Proof. The implications (3) $\Rightarrow$ (1) and (1) $\Rightarrow$ (2) are immediate. Thus it suffices to prove that (2) implies (3). We urge the reader to find their own proof of this fact. Let Z' and Z'' be closed subspaces with $T = |Z'| = |Z''|$ such that $Z' \to Y$ is a proper morphism of algebraic spaces. We have to show that $Z'' \to Y$ is proper too. Let $Z''' = Z' \cup Z''$ be the scheme theoretic union, see Morphisms of Spaces, Definition 0CYZ. Then Z''' is another closed subspace with $|Z'''| = T$. This follows for example from the description of scheme theoretic unions in Morphisms of Spaces, Lemma 0CZ1. Since $Z'' \to Z'''$ is a closed immersion it suffices to prove that $Z''' \to Y$ is proper (see Morphisms of Spaces, Lemmas 04XZ and 04XY). The morphism $Z' \to Z'''$ is a bijective closed immersion and in particular surjective and universally closed. Then the fact that $Z' \to Y$ is separated implies that $Z''' \to Y$ is separated, see Morphisms of Spaces, Lemma 05Z2. Moreover $Z''' \to Y$ is locally of finite type as $X \to Y$ is locally of finite type (Morphisms of Spaces, Lemmas 06ED and 03XG). Since $Z' \to Y$ is quasi-compact and $Z' \to Z'''$ is a universal homeomorphism we see that $Z''' \to Y$ is quasi-compact. Finally, since $Z' \to Y$ is universally closed, we see that the same thing is true for $Z''' \to Y$ by Morphisms of Spaces, Lemma 08AJ. This finishes the proof. $\square$